



Insulin-like growth factor 1 (IGF1) over-expression on a phycocyanin carrier protein in cyanobacteria

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Abstract

The Insulin-like Growth Factor 1 (IGF1) is a small (~7.6 kDa) protein-hormone primarily generated in the liver of mammals. It has a similar structure to insulin, and functions to support cell growth and development. As a result, IGF1 is in high demand in the biopharmaceutical, biomedical, and research fields. In eukaryotic photosynthetic systems, over-expression of this and other proteins has been met with limited success. The current study used *Synechocystis* sp. PCC 6803, a freshwater, single-celled cyanobacterium, as a host system to produce the IGF1 through recombinant DNA technology. To stabilize and enhance IGF1 accumulation in the cells, fusion constructs of the codon optimized *IGF1* gene with the β -subunit (*cpcB* gene) of phycocyanin were designed. The resulting transgenic strains accumulated significant amounts of recombinant IGF1 in direct proportion to phycocyanin in *Synechocystis*. Quantitative aspects of the work, including the culture biomass yield in photoautotrophic and mixotrophic growth conditions, and the amount of recombinant fusion protein (CpcB*IGF1), as a fraction of total cellular protein in *Synechocystis* are reported. Further, antibody specificity is used as a tool to examine the effect of different fusion orientations between the CpcB and IGF1 proteins, needed to inform of optimal conditions for IGF1 functional properties. The work adds to the concept of cyanobacteria serving as cell factories for the renewable photosynthetic generation of biopharmaceutical proteins from sunlight, carbon dioxide and water.

Keywords Fusion DNA and protein constructs · Phycocyanin · Recombinant protein stability · *Synechocystis* · Cyanobacterium

Introduction

Insulin-like Growth Factors (IGFs) are a class of small hormonal proteins that are primarily generated in the liver of mammals and aid in organismal cell growth and development (Maki 2010; Khan et al. 2025). Mammals, including humans and bovines, possess two different IGFs: IGF1 and IGF2 (Poreba and Durzynska 2020; LeRoith et al. 2021). The human and bovine *IGF1* genes encode identical 70 amino acid long polypeptides with a molecular weight of approximately 7.6 kDa (Vajdos et al. 2001; Romand et al. 2016; Rotwein 2017). The tertiary configuration of the IGF1 proteins (<https://www.uniprot.org/>, <https://alphafoldserver.com/>) comprise an asymmetrical helix-coil protein structure,

having a catalytic site closer to its N-terminus domain (Li et al. 2019).

Because of its importance in cell growth and development, and in addition to its use in basic research, IGF1 is in high demand as a growth supplement by the cultured meat and leather industries. This necessitates sufficient supplies to meet the growing demand. Photosynthetic organisms provide an interesting platform for recombinant IGF1 production as they do not need an external source of nutrients to grow (Panahi et al. 2003; Poudel et al. 2017; Yao and Asayama 2017; Park et al. 2020). In addition, the risk of culture contamination is significantly lower than that of a mammalian or fermentative cell system, because photosynthetic organisms rely solely on sunlight, CO₂ and simple inorganic nutrients. Cyanobacteria are of interest in this respect, as some strains can thrive under high alkalinity and salinity conditions, which further minimize chances of contamination by invading microorganisms (Chaves et al. 2015). However, producing recombinant IGF1 in higher plants is time-consuming, as plants grow

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slowly and require complex and often unstable genetic transformation protocols. Furthermore, the stability of recombinant proteins themselves in non-native organisms, especially plants and algae, is often severely compromised by intracellular enzymes and the cellular proteasome that recognize and degrade exogenous proteins (Zhang et al. 2021), resulting in very low steady state yields (Majhi and Melis 2025a). Such pitfalls can be alleviated using cyanobacteria.

Unicellular cyanobacteria offer a platform to produce a wide variety of bioactive chemicals (Lindberg et al. 2010; Singh et al. 2011, 2020; Formighieri and Melis 2016; Price et al. 2020; Santos-Merino et al. 2023) and recombinant proteins (Betterle et al. 2020; Hidalgo Martinez et al. 2022; Majhi and Melis 2025a). They grow much faster than plants and are amenable to stable genetic transformation, including spontaneous uptake of exogenous DNA and a direct double homologous recombination, leading to successful transformants, suitable for the expression of recombinant enzymes and other proteins (Berla et al. 2013). These properties make them ideal hosts for recombinant protein production. Furthermore, stability of recombinant protein expression in cyanobacteria is enhanced by the fusion of otherwise unstable recombinant proteins with highly expressed native proteins, e.g. phycocyanin, which has been shown to substantially improve recombinant protein stability and accumulation within the cells (Formighieri and Melis 2015, 2016; Zhang et al. 2021; Hidalgo Martinez and Melis 2023; Majhi and Melis 2024).

In the present study, the unicellular cyanobacterium *Synechocystis* sp. PCC 6803 (*Synechocystis*) was used as a model organism and a host to test for the expression of IGF1. Four genetic fusion constructs between the phycocyanin CpcB β -subunit and the *IGF1* gene were designed and successfully integrated into the *Synechocystis* genome for recombinant IGF1 production. Two of these constructs entailed a CpcB C-terminus fusion to the IGF1 N-terminus (forward IGF1 orientation). The other two entailed a CpcB C-terminus fusion to the IGF1 C-terminus (inverse IGF1 orientation). The fusion constructs of forward-IGF1 and inverse-IGF1 with the β -subunit of phycocyanin were successfully expressed and accumulated in stoichiometric amounts with the CpcB protein. The synthesized Phyco*IGF1 proteins (CpcB*IGF1 and CpcB*invIGF1) were effectively isolated from crude cellular extracts using differential cobalt affinity column chromatography. Moreover, the IGF1 protein was successfully cleaved and separated from the Phyco*IGF1 fusion form, upon the action of recombinant proteases. The current study also examined the effect of antibiotic and glucose on *Synechocystis* cell growth and biomass yield. It suggested a cost-effective and efficient method for using recombinant DNA technology to exploit cyanobacteria as photosynthetic cell factories in the production of IGF1.

Materials and methods

Genomic DNA constructs

Genetically modified cyanobacterial strains were generated upon transformation of the wild-type *Synechocystis*. The highly expressed phycocyanin *cpc* operon in *Synechocystis*, consisting of *cpcB*, *cpcA*, *cpcC2*, *cpcC1*, and *cpcD* genes that code for different subunits of the abundant phycocyanin (Fig. 1A) were the subject of this genetic transformation. The *cpc* operon was modified upon replacing the native *cpcB* gene, encoding the β -subunit of phycocyanin, with a fusion construct of the *cpcB* in the leading position and the *IGF1* in the trailing position of the linear construct. The latter was fused in the forward (N-to-C) or inverse (C-to-N) orientation (please see below). In this arrangement, the IGF1 gene was inserted prior to the stop codon at the carboxyl terminus of the *cpcB* gene. Additional DNA sequences were inserted between the *cpcB* and the *IGF1* fusion, including the 6xHis tag, a spacer (S: PAEKWAPGGS), the Tobacco Etch Virus (TEV) protease cleaving sequence (tev: ENLYFQ/G), followed by the *Synechocystis* codon optimized DNA sequence of the Insulin-Like Growth Factor 1 (IGF1). Upon double homologous recombination, the resulting fusion constructs *cpcB*6xHis*tev*IGF1* and *cpcB*6xHis*S*tev*IGF1* replaced the native *cpcB* gene in the *Synechocystis* genome (Fig. 1B and C). The transgenic strains harboring the aforementioned DNA constructs in the genome were termed *Syn*tev*IGF1* and *Syn*S*tev*IGF1* (Fig. 1B,C).

Two additional fusion constructs was generated by assembling DNA sequences of *cpcB*, a spacer (S: PAEKWAPGGS), 6xHis tag, the Human Rhinovirus 3C (HRV) protease cleaving sequence (hrv: LEVLFQ/GP), followed by the codon optimized DNA sequence of the Insulin-Like Growth Factor 1 (invIGF1). The resulting fusion constructs *cpcB*6xHis*hrv*invIGF1* and *cpcB*S*6xHis*hrv*invIGF1* replaced the native *cpcB* gene in the *Synechocystis* genome (Fig. 1D,E). The transgenic strains harboring the DNA constructs were termed *Syn*hrv*invIGF1* and *Syn*S*hrv*invIGF1* (Fig. 1D,E).

The spacer S was designed with the inclusion of two prolines to change the relative orientation of the IGF1 protein relative to that of the CpcB β -subunit and thus to probe for potential hindrances in the assembly of the $(\alpha,\beta\text{IGF1})_3\text{G1}$ heterohexameric disc (please see discussion section). The cleaving sequences tev and hrv were introduced in the constructs to test for the cleaving efficiency of the IGF1 from the isolated $(\alpha,\beta\text{IGF1})_3\text{G1}$ heterohexameric Phyco*IGF1 fusion protein forms. All the DNA constructs were designed in the lab and were synthesized by Biomatik

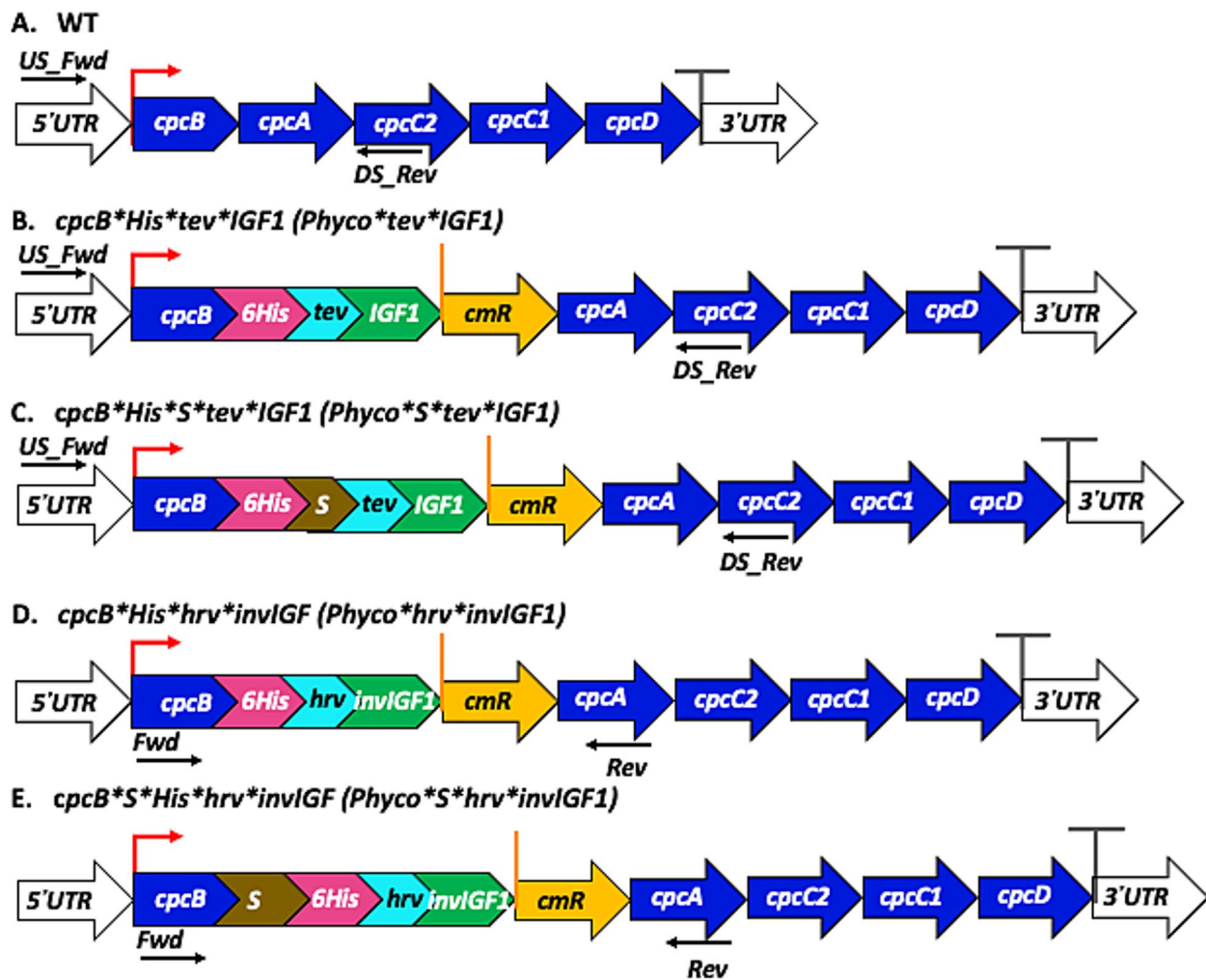


Fig. 1 Schematics of the native and modified *cpc* operon in *Synechocystis* wild-type (WT) and transformant strains. The 5' UTR contains the ribosome-binding sequence 5'-GTAGGAGATTAA TTCA-3', which is followed by the ATG start codon of the *cpcB* gene, denoted by the forward red arrow. The orange "I" denotes the presence of an exogenous intergenic sequence, introduced prior to the chloramphenicol resistance cassette (*cmR*), whereas "T" denotes the operon terminator at the end of the *cpcD* gene, followed by the 3' UTR of this operon. (A) The *Synechocystis* native *cpc* operon comprising phycocyanin-encoding *cpcB* and *cpcA* and linker polypeptides *cpcC2*, *cpcC1*, and *cpcD* genes presented in blue color and marked as WT. (B) The modified *cpc* operon encoding CpcB, 6xHis, tobacco etch virus protease cleaving site (*tev*: ENLYFQ/G), and *Synechocystis* codon-optimized bovine "Insulin-like Growth Fac-

tor 1" (*IGF1*) gene in fusion construct configuration, followed by a chloramphenicol resistance cassette (*cmR*) in a non-fusion operon configuration, denoted as *cpcB*6xHis*tev*IGF1* (*Phyco*tev*IGF1*). (C) Same as in (B) with the addition of a spacer (S: PAEKWAPGGS) between *6xHis* and *tev*, denoted as *cpcB*6xHis*S*tev*IGF1* (*Phyco*S*tev*IGF1*). (D) Same as in (B) upon replacement of the *tev* cleaving sequence with that of the human rhinovirus protease cleaving site (*hrv*: LEVLFQ/GP), denoted as *cpcB*6xHis*hrv*invIGF1* (*Phyco*hrv*invIGF1*). (E) Same as in (D) with the addition of a spacer (S: PAEKWAPGGS) between the *cpcB* and *6xHis*, denoted as *cpcB*S*6xHis*hrv*invIGF1* (*Phyco*S*hrv*invIGF1*). The location of forward and reverse primers is shown in black arrows. The nucleotide sequences of the transgenes are shown in the Supplementary Data section

USA (Wilmington, DE, USA). The resulting corresponding plasmids were used to transform wild type *Synechocystis* as described by Majhi and Melis (2024).

All cyanobacterial strains were maintained on BG-11 agar plates containing 1% agar, 0.3% sodium thiosulfate, 10 mM TES-NaOH (pH 8.2), and appropriate antibiotics (kanamycin, KanR, 25 µg mL⁻¹ or chloramphenicol, CmR,

15 µg mL⁻¹). All plates and cultures were kept at 25 °C under continuous illumination at 40 µmol photons m⁻² s⁻¹.

Genomic DNA PCR analysis

To determine the genomic DNA state of transformants, colony PCR analysis was performed using cyanobacterial liquid

cultures. A small aliquot of cyanobacterial culture (20 μL) was mixed with 100% ethanol (20 μL) and Chelex 100 resin (100 μL , 10% w/v) (BioRad: 142–1253) and heated at 98 °C for 10 min prior to centrifugation at 16,000 $\times g$ for 10 min. The supernatant was used as a DNA template for the PCR analysis, as follows. In a 25 μL reaction mixture, 2.5 μL of supernatant was combined with Q5 High-Fidelity 2X Master Mix (New England Biolab, USA) for the PCR reaction. The manufacturer's instructions were followed to accurately set the annealing temperature for primers *US Fwd*: 5'-ACC TGTAGAGAAGAGTCCC-3'; *DS_Rev*: 5'-GCGGAATAT TGTC AACCAG-3' (Fig. 1B,C), and *cpcB-Fwd*: GTATTC ACTCGGGTT; *cpcA-Rev*: GGTGCGGTTGATTTCATC (Fig. 1D,E). Amplified PCR products were analyzed on a 1% agarose gel to identify the DNA PCR products and thus to test the homoplasmy status of the transformants genomic DNA.

Cell growth and dry cell weight measurement

Cells were grown in 100 mL BG-11 liquid media until OD₇₃₀ reached 0.8–1.0. Cells were harvested by centrifugation at 8,000 $\times g$ for 15 min, the supernatant was discarded, and the pellet was resuspended in 5 mL of fresh BG-11. In photoautotrophic growth cultivation, cells were inoculated to an OD₇₃₀ of 0.05 in 100 mL of BG-11 in the absence or presence of antibiotic, as a selectable marker. In mixotrophic growth, cells were grown in the presence of 5 mM glucose in addition to the above-mentioned conditions. Cultures were kept on a shaker (rpm: 100) at 30 °C under continuous illumination (40 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$) for 7 days. The OD₇₃₀ was measured every 24 h. The rate of growth was plotted as a function of growth time, using the GraphPad Prism or KaleidaGraph software. Rate of cell growth was assessed from the steep portion of the growth curves. Following 7 days of growth, cells were pelleted by centrifugation at 8,000 $\times g$ for 15 min. Wet pellets were rinsed twice and dried at 90 °C for approximately 2–3 h prior to dry weight measurement. Dry cell weight data were plotted using the GraphPad Prism software.

Whole cell absorption spectra were measured as described (Majhi and Melis 2024). Cells were pelleted by centrifugation at 5,000 $\times g$ for 15 min after cultures reached an OD₇₃₀ of 0.8–1.0. Pellets were resuspended in aliquots of 2 mL of BG-11 prior to measurement. A Shimadzu UV-1800 spectrophotometer was used to measure the absorbance spectra, with the latter suspended at an optical cell density OD₇₃₀ = ~0.4. The optical path length and scanning speed were set to 1 cm and 200 nm min⁻¹, respectively.

Chlorophyll *a* concentration was measured as described in Kirst et al. (2014). Chlorophyll and carotenoid pigments were extracted from whole or broken cells using 100% methanol, as the solvent. The methanolic solution containing

these pigments was used to measure the absorbance spectra. The absorbance of the methanolic extract at 663 nm was used to calculate the chlorophyll *a* content of the samples. Similarly, the A₆₂₅ nm value from the whole cell absorbance spectra was used to estimate the phycocyanin content of cells.

Protein isolation

Cells were harvested at an OD₇₃₀ of 0.8–1.0 by centrifugation at 8,000 $\times g$ for 15 min. Pellets were resuspended in 25 mM Tris-HCl, pH-8.2, solution supplemented with cOmplete mini protease inhibitor cocktail (2 mL per 300 mL cell culture). The cell suspension was passed through a French Press three times at a pressure of 1,500 PSI to disrupt the cells. Unbroken cells were pelleted by centrifugation at 400 $\times g$ for 3 min. The 6xHis-tagged proteins were isolated from the crude cellular extracts using a differential cobalt-affinity column chromatography. Prior to the His-tagged protein isolation, the crude cellular extracts were incubated in 20 mM HEPES buffer, pH 7.5, containing 0.5% Triton-X 100 at 4 °C for 30 min in a spinning rotor (15 cycles min⁻¹). Following incubation, the crude cellular extracts were centrifuged at 16,000 $\times g$ for 5 min to remove insoluble materials. The supernatant, containing soluble proteins, were mixed with cobalt-resin (His-Select Cobalt Affinity Gel, Millipore Sigma) in a 1:1 (v:v) ratio and incubated at 4 °C for 30–45 min in a spinning rotor (15 cycles min⁻¹) to facilitate protein binding to the resin. The mixture was then passed through a column and washed four times with 20 mM HEPES, pH-7.5, 150 mM NaCl, and 10 mM imidazole washing solution (10 mL/wash) to remove unbound proteins. The His-tagged fusion proteins were eluted from the Co column using 20 mM HEPES, pH-7.5, 150 mM NaCl, and 250 mM imidazole elution solution. Eluted proteins were concentrated using Amicon Ultra 15 mL centrifugal filters (Millipore Sigma). The concentrated protein samples were snap frozen in liquid nitrogen and stored at -80 °C for later use.

SDS-PAGE and NATIVE-PAGE analysis

The crude cell extracts were used to resolve the total *Synechocystis* cellular proteins. 1 μg chlorophyll *a* equivalent of crude cell extract was solubilized in 1 $\times$ Laemmli Sample Buffer (Bio-Rad, USA) for 30 min at room temperature. To facilitate solubilization and protein denaturation, 1 M urea and 5% β -mercaptoethanol were added to the mix, and samples were gently vortexed every 10 min. Following incubation, samples were centrifuged at 16,000 $\times g$ for 3 min and the supernatant loaded onto a Precast 12-well kD Mini-PROTEAN TGX Protein Gel (Bio-Rad) and electrophoresed at 200 V for 45 min. For the SDS-PAGE and NATIVE-PAGE

analysis of cobalt column eluted protein samples, 20 µg of isolated heterohexameric phycocyanin complexes were loaded onto the gel lanes.

Zinc chromophore fluorescence and Coomassie staining

Zinc sulphate (ZnSO_4) solution was used to label the phycocyanobilin chromophore-binding proteins directly on the gel following the SDS-PAGE analysis, as follows. The electrophoresed gel was soaked in a 5 mM zinc sulphate solution for 30 min at room temperature with gentle shaking. A Chemidoc imaging system (Biorad) was used to capture the zinc-chromophore fluorescence of the phycocyanobilin-containing protein bands, with the labeling images seen under UV light.

The gel was subsequently stained in a Coomassie stain solution (0.07% brilliant blue R-250, 50% methanol, and 10% acetic acid) for 60 min at room temperature with gentle shaking to visualize and quantify specific protein bands of interest in this work. Coomassie stained gel images were taken using a Gel Doc XR + System and Image Lab software (Bio-Rad).

TEV and HRV3C cleaving of the IGF1 from its carrier protein complex

Cleaving the rIGF1 from the isolated $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ Phyco*IGF1 fusion constructs was performed as follows: 20 µg of isolated $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ harboring the Phyco*tev*IGF1 fusion protein complexes were incubated with variable concentrations (0, 20, 40, and 80 IU) of recombinant-TEV protease (rTEV) at 30 °C for 3 h. In case of Phyco*hrv*invIGF1 protein, 20 µg of isolated $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ heterohexameric complexes harboring the Phyco*hrv*invIGF1 fusion protein were incubated with 0 and 1 IU of recombinant-HRV 3C (rHRV) protease for 12–16 h only, at 4 °C. The shorter incubation in this case was chosen as a recent study (Majhi and Melis 2025b) showed HRV 3C to be a more efficient protease for cleaving recombinant proteins from the CpcB*P fusion constructs. Following incubation, protein samples were prepared for SDS-PAGE analysis, as described in the SDS-PAGE subsection above. Densitometry measurements of Coomassie-stained protein bands was performed using the built-in program of the Gel Doc XR + System (Bio-Rad).

Western blot analysis

20 µg of isolated $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ heterohexameric phycocyanin protein complexes were processed as described above for SDS-PAGE and NATIVE-PAGE analysis. Following the electrophoretic separation of the $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ constituent

proteins, proteins were transferred to a polyvinylidene fluoride (PVDF) membrane and probed with anti-CpcB and anti-IGF1 specific polyclonal antibodies as described (Majhi and Melis 2024, 2025a).

Results

Genomic DNA status of transformants

A description of the genomic DNA organization of the wild type and transformant *cpc* operons in *Synechocystis*, which encode the respective proteins of the phycobilisome phycocyanin peripheral rods, is given in the Materials and Methods section, under “**Genomic DNA constructs.**”

The genomic DNA status of transformants was evaluated to test for the attainment of homoplasmy, meaning removal of wild type copies of DNA (Fig. 1A) and quantitative replacement with the corresponding transgenic version harboring the heterologous fusion constructs (Fig. 1B–E). Attaining DNA copy homoplasmy is important for determining the expression efficiency of the heterologous proteins, and the physiological and biochemical traits of genetically modified strains.

A colony-PCR analysis was performed with cellular DNA as a template to test the genomic DNA status of cells. The primers used in the PCR analysis were designed to cover the modified region of the *cpc* operon in the genome, as well as the upstream and downstream native DNA regions of the transgenic DNA insertion. The PCR analysis of DNA from the wild type (WT), *Syn*S*tev*IGF1*, and *Syn*tev*IGF1*, showed a single product for each strain on agarose gel electrophoresis (Fig. 2A). The product sizes were 1,745 base pairs (bp), and 2,929 bp, and 2,893 bp, respectively. Absence of a PCR product corresponding to the size of WT in the transformants is evidence that these cells achieved DNA copy homoplasmy. Similarly, transgenic strains *Syn*hrv*invIGF1*, and *Syn*S*hrv*invIGF1* harboring inverse-IGF1 DNA constructs *cpcB*6xHis*hrv*invIGF1*, and *cpcB*S*6xHis*hrv*invIGF1*, respectively, exhibited a single PCR product on agarose gel electrophoresis (Fig. 2B). The product sizes of WT, *Syn*hrv*invIGF1*, and *Syn*S*hrv*invIGF1* were 983 bp, 2,140 bp, and 2,170 bp, respectively. The results show that all of the above transgenic strains have attained a state of DNA homoplasmy, meaning they possessed only transformant DNA and lacked the original wild type copies.

Photoautotrophic growth

Cell growth measurements are important to determine cell fitness and to assess the effect of genetic modifications on the cells' ability to conduct photosynthesis and growth. The

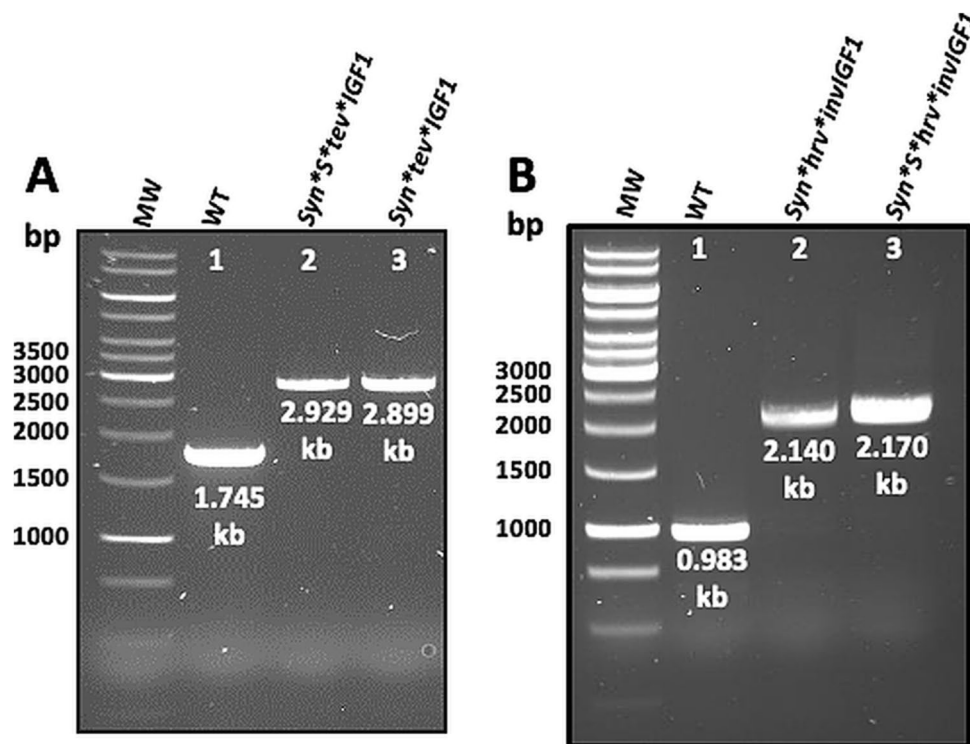


Fig. 2 Genomic DNA PCR analysis testing the *cpc* operon genotype of WT and *IGF1* transformants. A 1-kb DNA ladder (GeneRuler, ThermoScientific) was used as molecular weight (MW) markers (MW lane). Primers US_Fwd (5'-ACCTGTAGAGAAGAG TCCC-3'), and DS_Rev (5'-GCGGAATATTGTCAACCAG-3'), as shown in Fig. 1(A-C), and primers Fwd (GTATTCACCTCGGGTT), and Rev (GGTGCGGTTGATTTCATC), as shown in Fig. 1(D-E) flanking the insertion site were used to amplify the genomic DNA of the corresponding strains. (A) The WT strain yielded a single 1.745 kb PCR product, corresponding to the native *cpc* operon. The

*Syn*S*tev*IGF1* and *Syn*tev*IGF1* strains yielded single 2.929 kb and 2.899 kb PCR products, respectively, corresponding to the modified *cpc* operon DNA in these transformants (Fig. 1(A-C)). (B) The WT strain yielded a single 0.983 kb PCR product, corresponding to the native *cpc* operon, probed with shorter distance primers. The *Syn*hrv*invIGF1* and *Syn*S*hrv*invIGF1* strains yielded single 2.14 kb and 2.17 kb PCR products, respectively, corresponding to the modified *cpc* operon DNA in these transformants (Fig. 1(D-E)). Absence of wild type PCR products in the transformants is evidence of transgenic DNA copy homoplasmy

photoautotrophic growth analysis revealed that all cell types were able to use light for photosynthesis. However, the rate of growth was slower in the transformants than in the WT (Fig. 3). Photoautotrophic growth doubling time for the *Syn*tev*IGF1* and *Syn*S*tev*IGF1* strains (presence of chloramphenicol) was 38.5 ± 4.5 h, which is approximately double that of the WT strain (24 h, no chloramphenicol) (Fig. 3A). Furthermore, the transformant cultures had a more greenish coloration, compared with the WT blue green (Fig. 3B). The slower growth under sub-saturating illumination ($40 \mu\text{mol photons m}^{-2} \text{s}^{-1}$) and the altered coloration of the strains are attributed to partial loss of phycocyanin in the transformants, as compared to that in the WT, and total loss of phycocyanin in the Δcpc strain (please see absorbance spectra below). This partial loss of the phycocyanin light-harvesting antenna was shown to be the reason for the slower growth of similar transformants under the sub-saturating intensity of irradiance, a difference that was alleviated, when irradiance levels approached the saturation level (Kirst et al. 2014; Formighieri and Melis 2015). The notion

of a truncated phycocyanin antenna in the transformants resulting in slower rates of light absorption and slower photosynthesis and growth under these conditions is further supported by the even slower rate of growth of the Δcpc strain in which the entire *cpc* operon has been deleted and, thus, there is no phycocyanin (Fig. 3A, Δcpc) (Kirst et al. 2014).

The presence (+A) or absence (-A) of the antibiotic selectable marker (chloramphenicol, $15 \mu\text{g mL}^{-1}$) in the growth medium was investigated in this portion of the work. Photoautotrophic growth analysis of transgenic strains containing the inverse-*IGF1* DNA constructs in the absence of antibiotic exhibited a slower rate of growth compared to the WT (Fig. 3C). The doubling time for the WT was 24 h, whereas the doubling times of the *Syn*hrv*invIGF1* and *Syn*S*hrv*invIGF1* were closer to 27 ± 1 h for the *Syn*hrv*invIGF1*-A and *Syn*S*hrv*invIGF1*-A strains. Interestingly, the presence of the selectable marked antibiotic (chloramphenicol) in the growth medium of the transformants (Fig. 3C, *Syn*hrv*invIGF1* + A and *Syn*S*hrv*invIGF1* + A) caused a further slowdown in

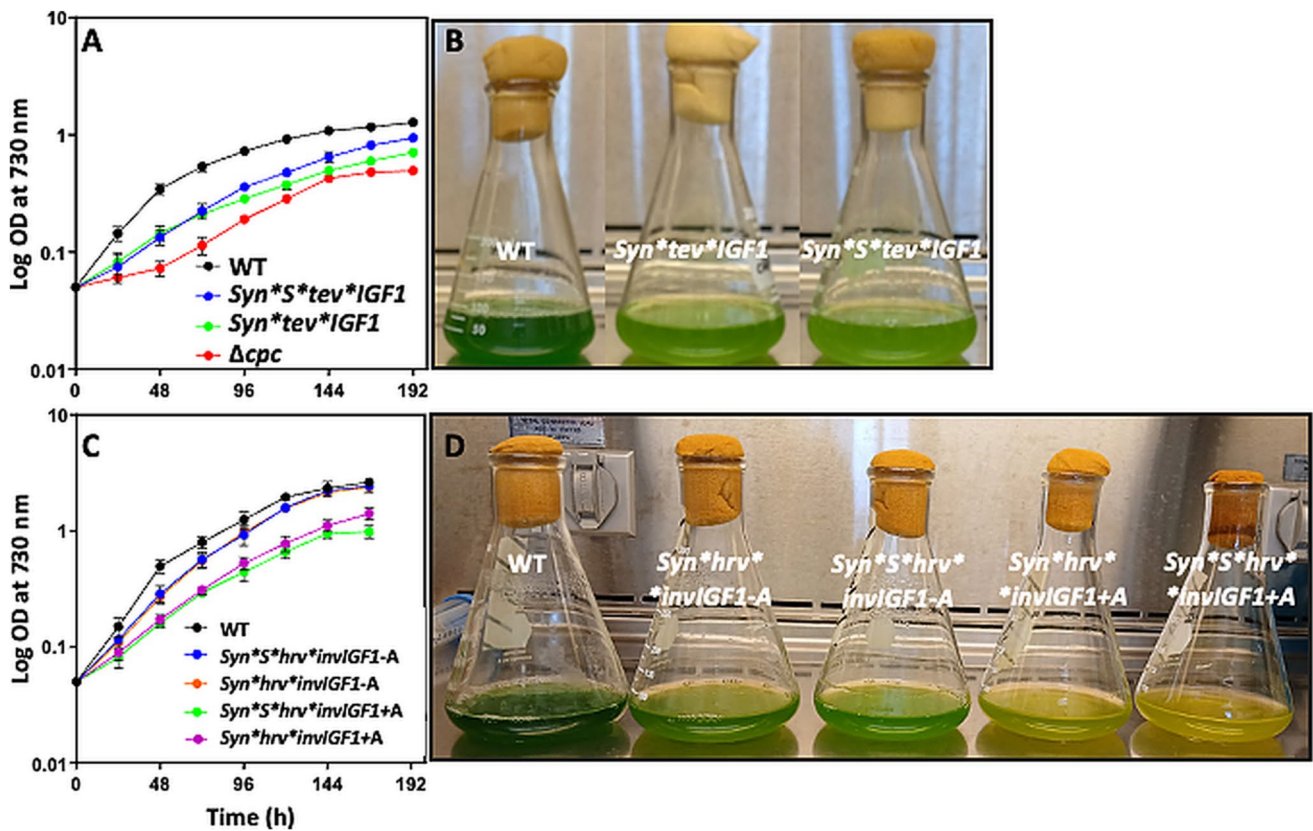


Fig. 3 Photoautotrophic growth curves and liquid culture coloration of *Synechocystis* WT and *IGF1* transformant strains in liquid culture. (A) Comparison of photoautotrophic growth kinetics in the absence of a selectable marker for the WT (black circles), but in the presence of selectable markers CmR for the *Syn*S*tev*IGF1* (blue), and *Syn*tev*IGF1* (green circles), and KanR for the Δcpc (red circles) strains. Plotted is the log OD at 730 nm of the cultures, as a func-

tion of growth time. (B) Coloration of the cells in BG-11 liquid media from the three strains highlighting the blue-green wild type and the more greenish *Syn*tev*IGF1* and *Syn*S*tev*IGF1* cultures. (C and D) Effect of the presence (+ A) or absence (-A) of antibiotic selectable marker (chloramphenicol $15 \mu\text{g mL}^{-1}$) on growth (C) and coloration (D) of transformants in liquid culture

growth to 37.5 ± 0.5 h (Fig. 3C). Slower photoautotrophic growth in the presence of the antibiotic chloramphenicol may be linked to the extra metabolic load for the processing and removal of chloramphenicol from the cells, and possibly due to adverse reactive oxygen species (ROS) production that cause damage to the photosynthetic apparatus. Chloramphenicol has been reported to promote ROS production under illumination conditions (Kodru et al. 2020). In this respect, cells growing in the presence of antibiotic appeared to have a lighter green coloration, compared to cells grown without antibiotic (Fig. 3D), consistent with the above-mentioned growth mitigating conditions.

Mixotrophic growth

Cell growth was also measured in the presence of 5 mM glucose, with or without antibiotic, to test for the effect of an external carbon source on cell doubling time. In this case, the presence glucose enhanced the rate of growth of all strains and mitigated the effect of antibiotic that was

observed under photoautotrophic conditions. (Fig. 4A). The doubling time for WT was lowered to 17 h, that of the *Syn*hrv*invIGF1 + A* and *Syn*S*hrv*invIGF1 + A* strains to 20.5 ± 0.5 h, whereas in the absence of antibiotic *Syn*hrv*invIGF1-A* and *Syn*S*hrv*invIGF1-A* strains exhibited a faster growth of 18.5 ± 0.5 h. Faster growth in the presence 5 mM glucose is linked to the supplementary cellular activities, supported in this case by both photosynthesis and glucose metabolism (Fig. 4B).

Biomass yield

Biomass accumulation is also an important variable in synthetic biology as it is linked to constitutive bioproducts yield. In this study, experiments were conducted to measure the dry cell weight (dcw) of strains employed, including WT and transgenics harboring the *invIGF1* protein in different fusion construct configurations. Under photoautotrophic growth conditions, WT cells accumulated approximately 0.66 g L^{-1} dcw (Fig. 5A) compared to 0.78 g L^{-1}

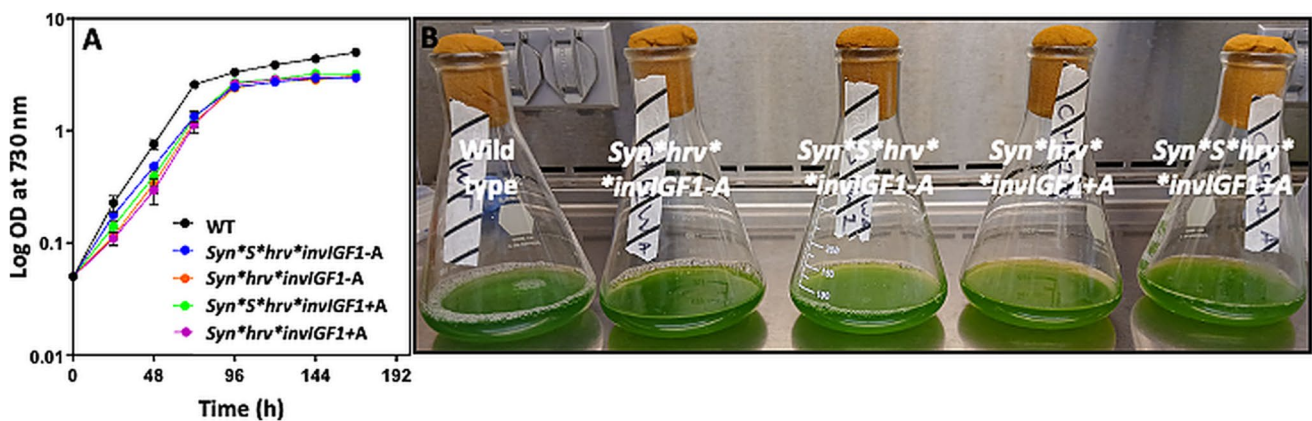
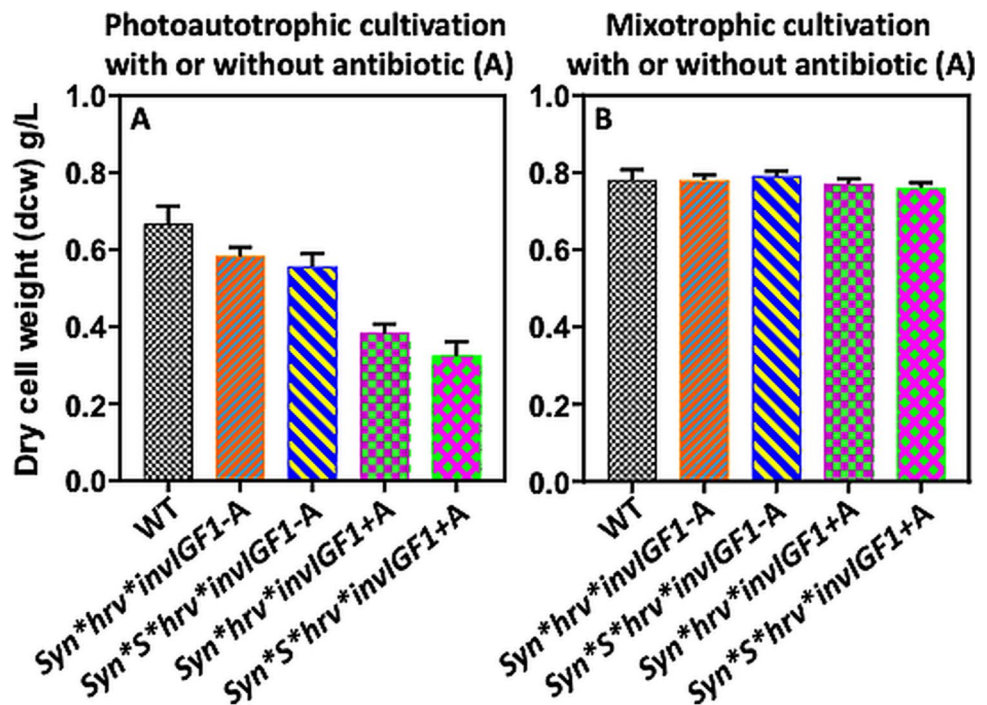


Fig. 4 Mixotrophic growth curves and liquid culture coloration of *Synechocystis* WT and *IGF1* transformant strains in the presence (+A) or absence (-A) of antibiotic selectable marker (chloramphenicol $15 \mu\text{g mL}^{-1}$) in liquid culture. BG-11 media were supplemented with 5 mM glucose in this experiment. (A) Comparison of photoautotrophic growth kinetics of the WT and transformants. Plotted is

the log OD at 730 nm of the cultures, as a function of growth time. (B) Coloration of the cells in BG-11 liquid media supplemented with 5 mM glucose. Presence of the latter helped to alleviate growth limitations and coloration differences seen under phototrophic growth in the presence (+A) or absence (-A) of chloramphenicol from the growth kinetics of the transformant strains

Fig. 5 Dry cell biomass yield of wild type and *IGF1* transformants under photoautotrophic (A) and mixotrophic (B) conditions in the presence (+A) or absence (-A) of $15 \mu\text{g mL}^{-1}$ chloramphenicol selectable marker. Biomass accumulation was measured seven days after culture inoculation, i.e., just prior to the onset of the stationary growth phase



dcw under mixotrophic conditions (Fig. 5B). Similarly, under photoautotrophic cultivation conditions, transgenic strains *Syn*hrv*invIGF1-A*, *Syn*S*hrv*invIGF1-A*, *Syn*hrv*invIGF1+A*, and *Syn*S*hrv*invIGF1+A*, accumulated 0.58, 0.55, 0.38, and 0.32 g L^{-1} dcw, respectively (Fig. 5A). Interestingly, under mixotrophic cultivation conditions such differences in were largely alleviated, resulting in greater biomass yield, on the average in the range of $0.775 \pm 0.015 \text{ g L}^{-1}$ dcw in all five strains examined (Fig. 5B).

Phycocyanin content

To determine the relative amount of phycocyanin assembled in cells under in vivo conditions, absorbance spectra of whole cell suspensions were measured. The absorbance spectrum of WT cells revealed four distinct bands with peaks at 440, 490, 625, and 680 nm (Fig. 6A,C). The absorbance peak at 440 nm is attributed to chlorophyll *a*, which is associated with photosystems I and II, whereas the chlorophyll-overlapping band at 490 nm is attributed to cellular

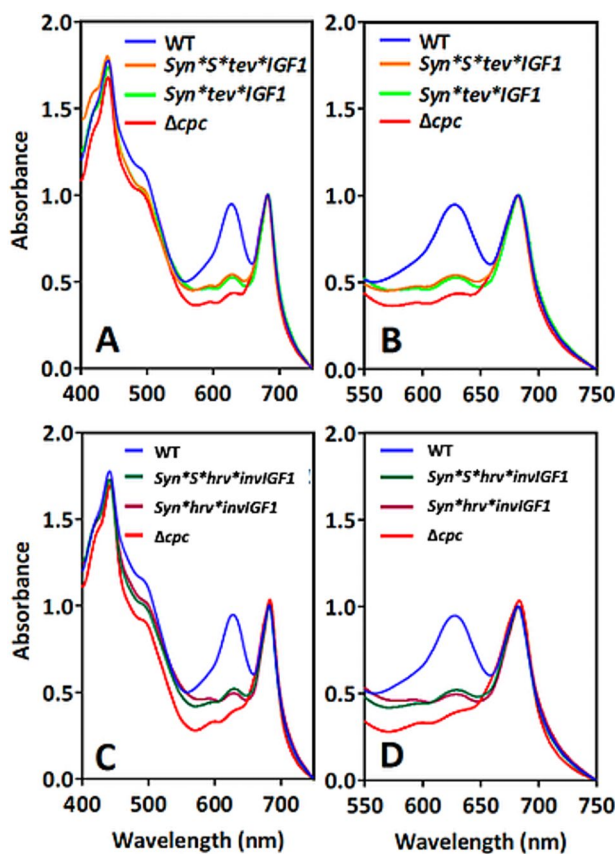


Fig. 6 In vivo absorbance spectra of intact *Synechocystis* WT and *IGF1* transformant strains probing for the presence of various light-harvesting pigments in the cells. All absorbance spectra were normalized to the Chl *a* absorbance max at 680 nm (set equal to 1.0) to enable phycocyanin pigment content comparisons. (A) The absorbance spectra of WT, *Syn*S*tev*IGF1*, *Syn*tev*IGF1*, and Δcpc strains are shown in blue, orange, and green, and red colors, respectively. (B) The absorbance spectra of WT, *Syn*S*hrv*invIGF1*, *Syn*hrv*invIGF1*, and Δcpc strains are shown in blue, green, purple, and red colors, respectively. Note the diminished phycocyanin content (A_{625} nm) in the transformants, relative to that in the wild type

carotenoids. The 625 and 680 nm bands (shown in greater resolution in Fig. 6B,D) are attributed to phycocyanin and chlorophyll *a*, respectively. The absorbance spectrum on the phycocyanin-lacking Δcpc strain, in which the entire *cpc* operon has been deleted (Kirst et al. 2014), is also shown. In the Δcpc strain, dominant are the chlorophyll *a* and carotenoid pigments, with the former showing a distinct 680 nm band and a lower amplitude chlorophyll *a* satellite peak at 625 nm. All four absorbance spectra have been normalized to the absorbance maximum of chlorophyll *a* at 680 nm to enable comparison in the cellular composition of phycocyanin among the four strains.

The absorbance spectra of transformants *Syn*tev*IGF1*, *Syn*S*tev*IGF1* and *Syn*hrv*invIGF1*, *Syn*S*hrv*invIGF1* exhibited a lower amplitude peak at 625 nm, corresponding to phycocyanin, compared to that

of WT, showing a lower phycocyanin content in these cells (Fig. 6). With the A_{625} of the Δcpc strain as baseline, we calculated the phycocyanin content in the *Syn*tev*IGF1*, *Syn*S*tev*IGF1* transformants to be $20.1 \pm 2.0\%$ of that in the WT (Fig. 6B). Similarly, the phycocyanin content in the *Syn*hrv*invIGF1*, *Syn*S*hrv*invIGF1* strains was estimated to be equal to $17.9\% \pm 2.0\%$ of that in the WT (Fig. 6D). These results are consistent with similar findings by Hidalgo Martinez et al. 2022, suggesting the assembly of only the proximal to the core disc of phycocyanin, with a modified the CpcB*IGF1 β -subunit, and absence of the middle and distant phycocyanin rod discs. Thus, the modified CpcB*IGF1 β -subunit with the attendant fusion construct must have caused the assembly of a truncated phycocyanin rod.

Total cell protein analysis and fusion protein isolation

Total cellular protein analysis was performed to test for the presence of Phyco*IGF1 fusion proteins in the various transformants. Crude cellular extracts were analyzed through SDS-PAGE Coomassie stain and zinc-phycobilin chromophore fluorescence measurements. Coomassie stain results for WT exhibited three main protein bands migrating to 55, 19, and 17 kDa corresponding to the large subunit of Rubisco (RbcL), β -subunit (CpcB), and α -subunit (CpcA) of phycocyanin, alongside other cellular proteins (Fig. 7A). In all transgenic strains (*Syn*tev*IGF1*, *Syn*S*tev*IGF1*, *Syn*hrv*invIGF1*, and *Syn*S*hrv*invIGF1*), protein bands corresponding to RbcL and CpcA were still observed. However, the amount CpcA was lower in the transgenics compared to WT. Furthermore, all transgenic strains lacked the 19 kDa CpcB protein band. Instead, a newly formed protein band migrating to ~26–28 kDa was observed. The latter originated from the CpcB*IGF1 fusion proteins, a conclusion supported by the pronounced zinc-phycobilin fluorescence labeling (Fig. 7B). Additionally, a 23 kDa protein band was observed in the protein extracts of transgenic strains, corresponding to the chloramphenicol antibiotic cassette (CmR), which was introduced as a selection marker.

Isolation of $(\alpha,\beta\text{*IGF1})_3\text{G1}$ heterohexameric complexes and cleaving of IGF1 from the CpcB*IGF1 fusion

To cleave the natural form of the IGF1 from the fusion proteins, recombinant constructs were designed to include a cleaving sequence (*tev*: ENLYFQ/G) specific for the Tobacco Etch Virus (TEV) protease, or (*hrv*: LEVLFQ/GP) human rhinovirus 3C protease, placed immediately upstream from the *IGF1* DNA sequence (Fig. 1B–E). His-tagged fusion proteins were isolated through

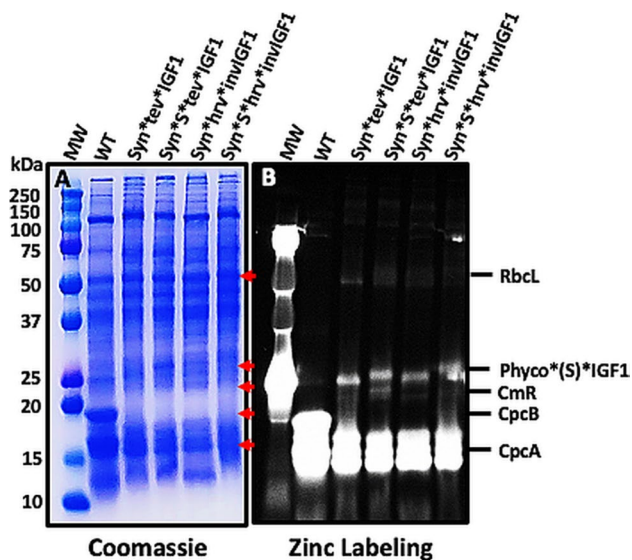


Fig. 7 SDS-PAGE analysis of total cell protein extracts from *Synechocystis* WT and *IGF1* transformant strains. **(A)** WT lanes: Coomassie-stained SDS-PAGE showing the protein profile of wild type cells. Based on Coomassie stain, the most abundant cellular proteins are the RbcL, CpcB, and CpcA. All transformants examined lacked the CpcB β -subunit band from the ~19 kDa position. Instead, they showed new protein bands in the 26–28 kDa regions. **(B)** Based on Zn-phycobilin chromophore fluorescence, protein bands migrating to 26–28 kDa were assigned to the Phyco*(S)*IGF1 fusion proteins. 1- μ g Chl *a* equivalent was loaded on the lanes for the SDS-PAGE analysis

differential cobalt-affinity column chromatography from the crude cellular extracts as heterohexameric phycocyanin ($(\alpha, \beta^*IGF1)_3G1$) discs (Hidalgo Martinez et al. 2022; Hidalgo Martinez and Melis 2023), in which α is the α -subunit of phycocyanin, β^*IGF1 is the CpcB*His*(S)*tev*IGF1, or CpcB*(S)*His*hrv*invIGF1 fusion protein, and CpcG1 is the proximal phycocyanin disc linker that anchors this phycocyanin disc to the allophycocyanin core cylinders (Hidalgo Martinez et al. 2022; Hidalgo Martinez and Melis 2023). The isolated heterohexameric complexes served as templates for the IGF1 cleaving reactions. In the case of the $(\alpha, \beta^*tev*IGF1)_3G1$, the cleaving reaction mixture involved incubating a fixed amount of isolated $(\alpha, \beta^*tev*IGF1)_3G1$ protein complexes (20 μ g protein) with different concentrations (0–80 IU) of commercially available recombinant rTEV protease at 30 °C for 3 h. Following incubation, reaction mixture samples were analyzed by SDS-PAGE to determine the cleaving and separation efficiency of the IGF1 protein from the fusion protein complexes.

Differential Co-affinity chromatography isolates and their analysis by SDS-PAGE from the Syn*S*tev*IGF1 strain showed three main protein bands migrating to 29, 27, and 17 kDa, corresponding to the linker CpcG1, Phyco*IGF1, and CpcA polypeptides, respectively (Fig. 8A, 0 IU). Addition of rTEV (20, 40, and 80 IU) to the reaction mixture

harboring the $(\alpha, \beta^*IGF1)_3G1$ complex resulted in the progressively greater appearance of cleaved Phyco* (20.7 kDa) and free IGF1 (~10 kDa) protein bands (Fig. 8A,B). Protease cleaving efficiency was determined from the progressive decrease in the amount of the 27 kDa Phyco*IGF1 fusion protein and the concomitant appearance of the cleaved ~20.7 kDa Phyco* moiety that was observed as a function of the increasing amount of rTEV employed (Fig. 8B). Densitometry analysis of Coomassie-stained protein gels showed that the fusion protein Phyco*IGF1 decreased to 40%, 38%, and 25% of the original in the presence of 20, 40, and 80 IU of rTEV protease, respectively. Correspondingly, the amount of the cleaved Phyco* product was measured to be 60%, 70%, and 75% of the original Phyco*IGF1 in the presence of 20, 40, and 80 IU of rTEV protease, respectively (Fig. 9A).

Similarly, when $(\alpha, \beta^*hrv*invIGF1)_3G1$ heterohexameric phycocyanin complexes harboring the hrv cleaving sequence were incubated in the presence of 0 IU and 1 IU of HRV protease at 4 °C for 16 h, a greater cleaving efficiency was observed on Coomassie stained SDS-PAGE. Protein samples without HRV-treatment showed three main protein bands migrating to 29, 27 and 17 kDa corresponding to CpcG1, Phyco*IGF1, and CpcA proteins (Fig. 8C and D). The HRV-treated sample showed, in addition to Phyco*IGF1 (27 kDa) and CpcA (17 kDa), two new protein bands migrating to 24 kDa, originating from the added recombinant HRV, and ~21 kDa, originating from the cleaved Phyco*IGF1. There was also a protein band migrating to ~8 kDa, attributed to cleaved IGF1. However, IGF1 is a small protein of about 7.6 kDa giving rise to a faint Coomassie-stained band. The latter is sometimes hard to see, as destaining to visualize larger molecular mass proteins easily removes most, if not all, of the Coomassie from the freed IGF1. Densitometry analysis of protein bands from the Coomassie stained SDS-PAGE (Fig. 8C), showed that incubation of the isolated $(\alpha, \beta^*hrv*invIGF1)_3G1$ heterohexameric phycocyanin complexes, harboring the hrv cleaving sequence, with 1 IU of HRV protease caused a decrease of Phyco*hrv*invIGF1 proteins to about 12% of the original. Conversely, the cleaved Phyco* product accounted for about 88% of the original (Fig. 9B).

A side-by-side analysis of SDS-PAGE Coomassie stain, Zn-phycobilin chromophore fluorescence, and Western blot analysis was undertaken to probe for the presence and location of Phyco*IGF1 and the cleaved IGF1 in the protease treated protein samples (Fig. 10). The combined analysis revolved around the HRV reaction mixture proteins Phyco*IGF1 (~27 kDa), rHRV (~24 kDa), Phyco* (~21 kDa), CpcA (~17 kDa), and IGF1 (~7.6 kDa), all of which were detected in the Coomassie stain (Fig. 10A). Three of these protein bands (Phyco*IGF1, Phyco*, and CpcA) were detected with the Zn-phycobilin chromophore

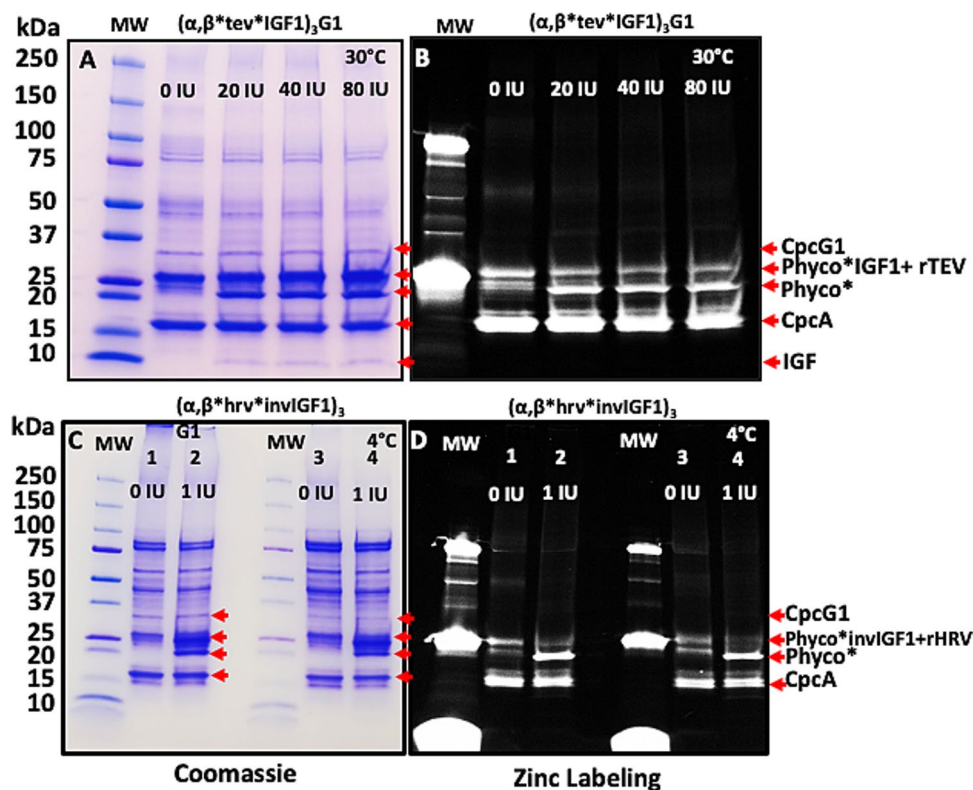


Fig. 8 Cobalt-column differential affinity chromatography, isolation of the $(\alpha,\beta^*\text{IGF1})_3\text{G1}$ fusion protein complexes, and rTEV/rHRV protease cleaving of the IGF1 protein from the Phyco*IGF1-associated protein complex. **(A)** SDS-PAGE and Coomassie-stained resolution of the $(\alpha,\beta^*\text{tev*IGF1})_3\text{G1}$ constituent proteins. (0 IU) Shown is the presence of the 29 kDa CpcG1 linker, the 27 kDa Phyco*IGF1, and the 17 kDa CpcA α -subunit of phycocyanin. Incubation of the isolated $(\alpha,\beta^*\text{tev*IGF1})_3\text{G1}$ proteins with 20, 40, and 80 IU of rTEV (GenScript, Piscataway, NJ) at 30 °C for 3 h showed the graduate release of the IGF1 from the heterohexameric complex, migrating to about the 7.6 kDa electrophoretic mobility position, and the concomitant appearance of a 21 kDa protein band, resulting from the de novo formation of CpcB*His*tev/Cpc*His*S*tev (marked as Phyco*) cleaving reaction products. **(B)** Zinc-chromophore fluorescence of phycocyanobilin proteins present in the samples of (A),

showing loss of fluorescence intensity from the 27 kDa protein band (0 IU) and gain in this signal at the 21 kDa position, emanating from the appearance of the CpcB*His*S*tev (Phyco*) proteolysis fragment. **(C)** Same as in (A), with the exemption of showing results with the isolated $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ proteins. Incubation of the isolated $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ proteins with 1 IU of rHRV at 4 °C for 12 h was sufficient to bring the cleaving reaction to quantitative completion. **(D)** Zinc-chromophore fluorescence of phycocyanobilin proteins occurring in the samples of (C), showing loss of fluorescence intensity from the 27 kDa protein band (0 IU) and gain in this signal at the 21 kDa position (1 IU), emanating from the appearance of the CpcB*S*His*hrv (Phyco*) proteolysis fragment. 20 μg of isolated fusion protein complexes were used in the protease cleaving reactions

fluorescence method (Fig. 10B), and only two (Phyco*IGF1 and IGF1) cross reacted in the Western blot analysis, conducted with specific anti-IGF1 polyclonal antibodies (Fig. 10C).

Differential antibody access to the $(\alpha,\beta^*\text{tev*IGF1})_3\text{G1}$ and $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ complexes

Specific polyclonal antibodies raised against the CpcB (anti-CpcB) and IGF1 (anti-IGF1) were employed to probe the stereochemistry and peripheral surface properties of the $(\alpha,\beta^*\text{tev*IGF1})_3\text{G1}$ and $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ heterohexameric phycocyanin complexes. Native-PAGE and

immunoblot analysis results of the $(\alpha,\beta^*\text{tev*IGF1})_3\text{G1}$ and $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ complexes are shown in Fig. 11. These undissociated complexes migrated to about 150 kDa, consistent with their protein composition. The immunoblot analysis showed that both anti-CpcB and anti-IGF1 strongly cross-reacted with the $(\alpha,\beta^*\text{tev*IGF1})_3\text{G1}$ complex. Surprisingly, these antibodies failed to cross-react with the $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ complex, suggesting a substantially different stereochemistry at the periphery of this complex, one that averted antibody access to the antigenic epitopes of either the CpcB or invIGF1 proteins. This is interpreted to be a consequence of the C-to-C configuration and the resulting difference in the folding of the invIGF1 in the $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ complex. In support, IGF1

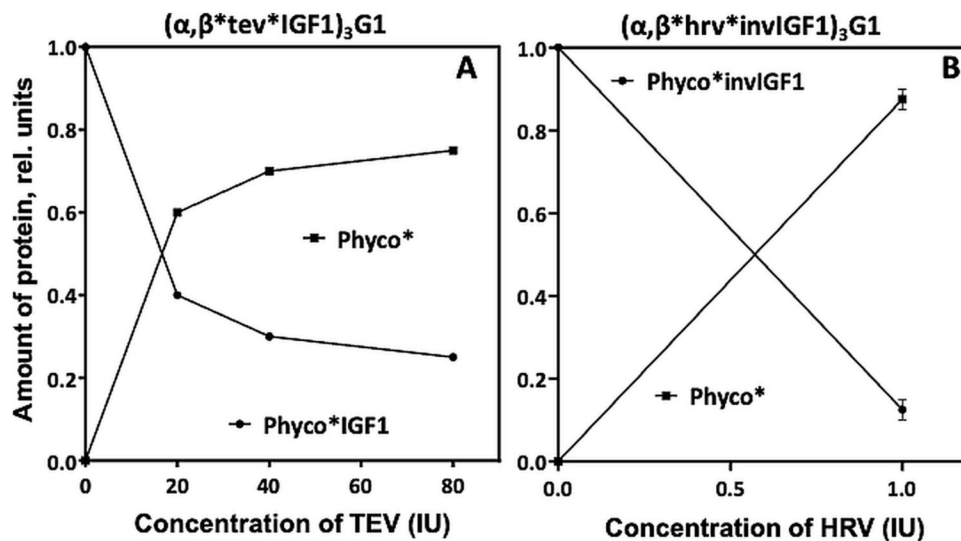


Fig. 9 Densitometric analysis of protein bands from the SDS-PAGE profile shown in Fig. 8. (A) Cleaving of the fusion protein Phyco*tev*IGF1 in the presence of rTEV at 30°C for 3 h, and accumulation of the reaction product Phyco*tev. (B) Cleaving of the fusion protein Phyco*hrv*invIGF1 in the presence of rHRV at 4°C for 12 h, and accumulation of the reaction product Phyco*hrv.

Amounts of Phyco*tev/hrv*invIGF1 reactants, and Phyco*tev/hrv products are plotted as a function of the rTEV/rHRV concentration used. Bio-Rad Image Lab software (Hercules, CA) was used for the densitometric analysis. Results are the average of four measurements with $(\alpha, \beta^* \text{tev/hrv}^* \text{IGF1})_3 \text{G1}$ fusion protein samples

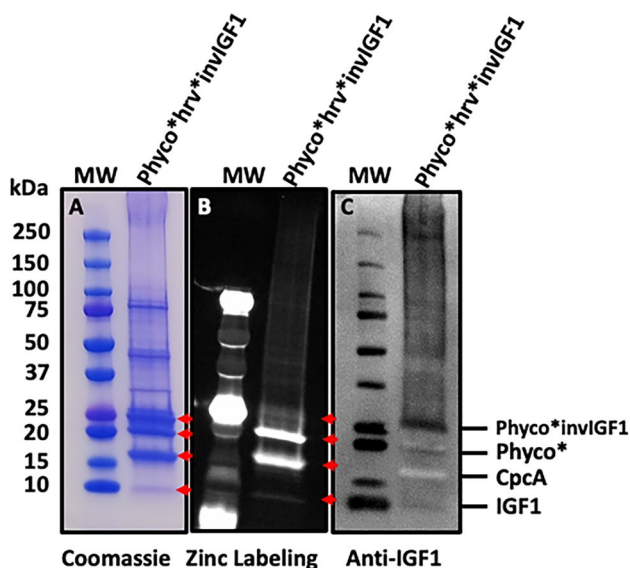


Fig. 10 (A) SDS-PAGE Coomassie stain, (B) Zn-chromophore labeling, and (C) Western blot analysis of 20 μg of isolated $(\alpha, \beta^* \text{hrv}^* \text{invIGF1})_3 \text{G1}$ fusion protein complexes incubated with 1 IU of rHRV protease at 4 °C for 12 h. The constituent proteins were separated by SDS-PAGE and probed with IGF1-specific polyclonal antibodies. A molecular marker is presented on the gel's left side for protein molecular weight estimation

and invIGF1 folding models, conducted through the Unit-Prot and AlphaFold protein modeling prediction software, showed that, despite the amino acid sequence identity, the

folding patterns of the IGF1 and invIGF1 are rather dissimilar (Supplementary Materials Fig. S1). The underlying reason for this dissimilarity is the way the peptide bonds form between different amino acids in the IGF1 and invIGF1 proteins (Lacroix et al. 1998; Kefala et al. 2021) (please see Supplementary Materials, page 6). It is concluded that an inverse IGF1 C-to-C fusion with the CpcB protein in the $(\alpha, \beta^* \text{hrv}^* \text{invIGF1})_3 \text{G1}$ heterohexamer contributed to structural properties that underlined the differential immunoblot results in Fig. 11 (please see discussion below).

Discussion

Structural aspects of phycocyanin

Phycobilisomes are the major pigment-protein complexes found in cyanobacteria, and they are an important component of the photosynthetic light harvesting system in these microorganisms (Singh et al. 2015; Zheng et al. 2021). They comprise large, multimeric, water-soluble protein complexes peripherally associated with the cytoplasmic side of the thylakoid membrane, where they are bound to the transmembrane CP43/CP47 chlorophyll-proteins of photosystem II (Kirst et al. 2014; Domínguez-Martín et al. 2022). Functionally, the phycobilisome plays an important role in the light-reactions of photosynthesis, capturing light energy and transferring the excitation energy to photosystem II, thereby

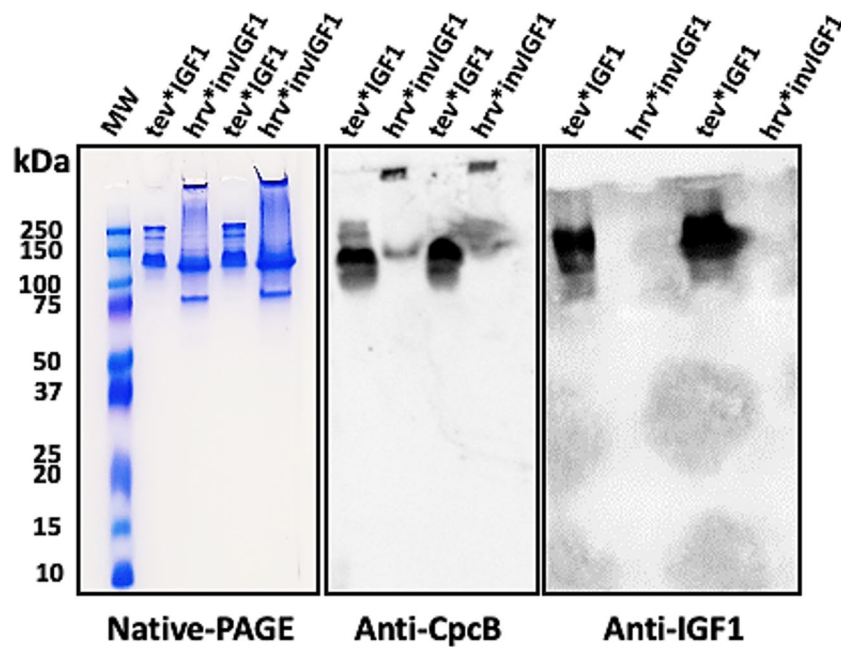


Fig. 11 Native-PAGE and immunoblot analysis of eluted $(\alpha,\beta^*\text{tev*IGF1})_3\text{G1}$ and $(\alpha,\beta^*\text{hrv*invIGF1})_3\text{G1}$ heterohexameric complexes harboring the two different CpcB*IGF1 fusion constructs. Left panel shows a Coomassie stain of the respective Native-PAGE analysis proteins. Electrophoretic mobilities to about 150 kDa are consistent with the calculated molecular weights of these heterohexa-

meric complexes. Middle panel shows an immunoblot analysis of the bands shown on the left panel with anti-CpcB antibodies. Right panel shows an immunoblot analysis with anti-IGF1 antibodies. These results showed that antigenic epitopes on CpcB and IGF1 can be inaccessible due to the conformation of the IGF1 in relation to the α,β heterohexamer

enhancing water (H_2O) oxidation and electron transport in the thylakoid membrane.

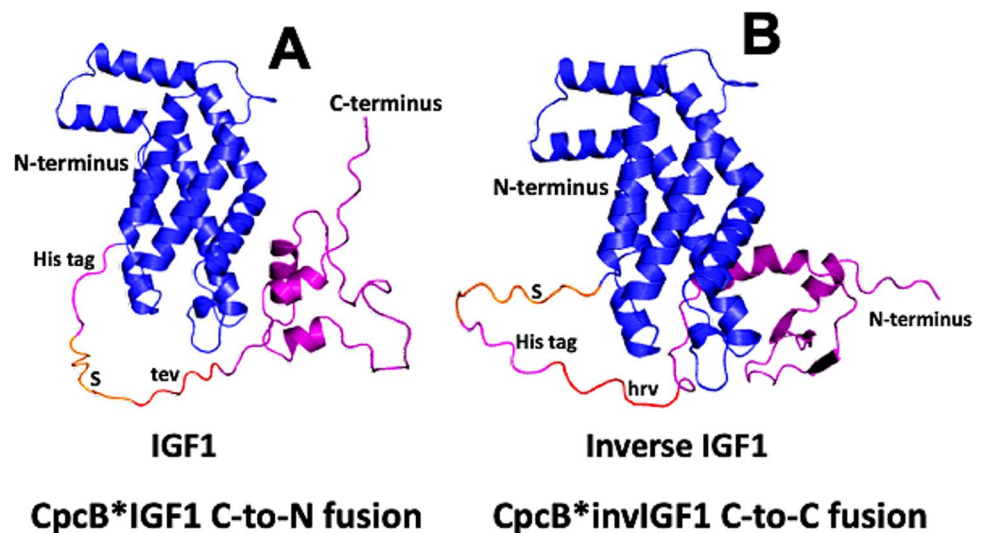
Phycobilisomes in *Synechocystis* possess three allophycocyanin core cylinders, and six phycocyanin-containing peripheral rods (Kirst et al. 2014; Domínguez-Martín et al. 2022). Each of the peripheral rods comprises 6 heterohexameric $(\alpha,\beta)_3$ discs, consisting of three α -CpcA and three β -CpcB subunits stacked to form the phycocyanin rods, and stabilized through their association with colorless linker polypeptides (Singh et al. 2015; Zheng et al. 2021). The peripheral rods are linked to the core cylinders via the colorless CpcG1 linker polypeptide, which is a structural component of the proximal to allophycocyanin $(\alpha,\beta)_3$ disc. The CpcG1 linker polypeptide is required for the structural and functional association of phycocyanin to allophycocyanin (Domínguez-Martín et al. 2022; Hidalgo Martinez et al. 2022; Hidalgo Martinez and Melis 2023). The CpcB and CpcA subunits of phycocyanin are the most abundant proteins in cyanobacteria (Betterle et al. 2020; Hidalgo Martinez and Melis 2023).

IGF1 vs invIGF1 and the CpcB*IGF1 and CpcB*invIGF1 fusion folding models

The *IGF1* gene encodes a protein of 70 amino acids, which has a folded helix-loop secondary structure (Nickle et al.

2024). The mature IGF1 protein shows a non-folding short N-terminal sequence, three core α -helices and their connecting loops, and an extended C-terminal loop (Ornitz and Itoh 2001) (Supplementary Materials, Fig. S1, left panel). In mammal and human physiology, the IGF1 protein binds onto the receptor IGF1R protein through multiple interacting residues, which are primarily distributed on the N-terminal, core helix of IGF1 (Li et al. 2019). It has been reported that the N-terminal residues of IGF1 are important for proper receptor binding and for eliciting the growth signal. With this in mind, we synthesized two different fusion-constructs resulting in the stable expression of the IGF1 protein. Fusion construct (Phyco*tev*IGF1) was designed to link the CpcB C-terminus to the IGF1 N-terminus (Fig. 12A), thereby exposing the IGF1 C-terminus to the medium in the periphery of the $(\alpha,\beta^*\text{IGF1})_3\text{G1}$ heterohexameric phycocyanin disc. Phyco*hrv*invIGF1 was designed to link the CpcB C-terminus to the IGF1 C-terminus (Fig. 12B), thereby exposing the IGF1 N-terminus to the medium in the periphery of the $(\alpha,\beta^*\text{invIGF1})_3\text{G1}$ heterohexameric phycocyanin disc. Orientation and amino acid sequence of the two constructs is shown in greater detail in the Supplementary materials, page 6. The folding models of Fig. 12 support the notion of a substantially different orientation and folding structure for the IGF1 versus the invIGF1 in relation to the leading CpcB protein.

Fig. 12 Folding models of the CpcB*IGF1 and CpcB*invIGF1. (A) CpcB C-terminus to IGF1 N-terminus fusion and folding model. (B) CpcB C-terminus to IGF1 C-terminus fusion and folding model. Orientation, placement and structure (tertiary configuration) of the invIGF1 in (B) is substantially different from that of the IGF1 in (A)



Protein folding patterns of the $(\alpha, \beta^* \text{tev}^* \text{IGF1})_3 \text{G1}$ and $(\alpha, \beta^* \text{hrv}^* \text{invIGF1})_3 \text{G1}$ complexes were obtained upon use of the UniProt (<https://www.uniprot.org/>), AlphaFold (<https://alphafoldserver.com/>) and of the $(\alpha, \beta)_3$ heterohexameric disk, adapted from Tan et al. (2018) to showcase folding patterns of the modified heterohexamers. In strains harboring the Phyco*IGF1 (C-to-N fusion; Fig. 13A) or Phyco*IGF1 (C-to-C fusion, Fig. 13B) constructs, we deemed the IGF1 protein to be localized peripherally and away from the center of the assembled $(\alpha, \beta)_3$ discs, supported by the functional association of the $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ discs with the allophycocyanin core cylinders, and from the stability of IGF1 in

these transformants (Hidalgo et al. 2022). This configuration allowed the rTEV and rHRV proteases to access the cleaving tev (ENLYFQ/G) and hrv (LEVLFFQ/GP) sequences, respectively, resulting in the release of the IGF1 protein in the reaction medium upon its separation from the Phyco*IGF1 fusion (Fig. 8). Additional support for the cytosolic exposure of the fusion construct to the medium is derived from the functional association of the His-tag with cobalt in the differential cobalt $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ disc association and release upon treatment with imidazole.

However, structural and functional differences existed between the $(\alpha, \beta^* \text{tev}^* \text{IGF1})_3 \text{G1}$ and $(\alpha, \beta^* \text{hrv}^* \text{invIGF1})_3 \text{G1}$

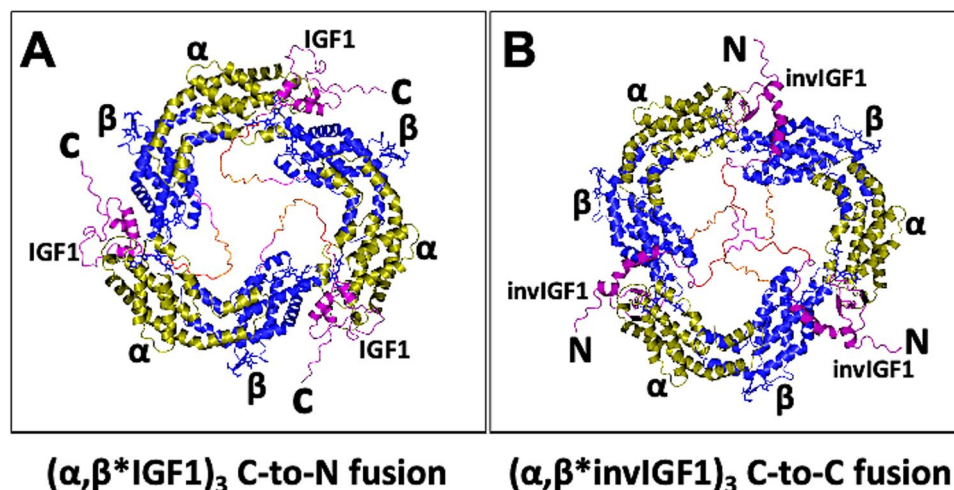


Fig. 13 The IGF1 fusion constructs assembled as $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ modified phycocyanin discs, containing the native CpcA α - and modified CpcB β -subunits of phycocyanin. (A) CpcB and IGF1 are shown in the C-to-N fusion orientation. (B) CpcB and invIGF1 are shown in the C-to-C fusion orientation. These $(\alpha, \beta^* \text{IGF1})_3 \text{G1}$ modified phycocyanin discs are functionally associated with the allophycocyanin core cylinders (Hidalgo et al. 2022), thus conferring stability

to the recombinant protein, while absorbing irradiance and transferring the excitation energy to the reaction center. CpcA α -subunits, CpcB β -subunits, and IGF1 are shown in olive-green, blue, and purple colors, respectively. The CpcG1 linker protein is omitted from the schematic for clarity of presentation. Note the substantially different topography (orientation and folding) of the IGF1 and invIGF1 in the periphery of the $(\alpha, \beta)_3$ heterohexameric complex

complexes, specifically as these pertain to anti-CpcB and anti-IGF1 antibody recognition of the respective epitopes in the two complexes. Both anti-CpcB and anti-IGF1 failed to recognize the respective targets in the $(\alpha, \beta^*hrv*invIGF1)_3G1$ complex (Fig. 11), suggesting that localization and orientation of the invIGF1 in the $(\alpha, \beta^*hrv*invIGF1)_3G1$ complex (Fig. 13B) masked the CpcB protein in a manner averting recognition by the specific anti-CpcB antibodies. These results suggested caution to be exercised, when phycocyanin fusion constructs are designed, to take into consideration the alignment of the target protein in relation to the (α, β) heterohexameric carrier disc, and the folding pattern of the “forward” versus “inverse” sequences.

The above are important considerations in synthetic biology, where high amounts of recombinant proteins properly folded for use in the biopharmaceutical sector, or catalytically active enzymes are required for enhanced pathway yield toward the synthesis of bioactive compounds and proteins. The plant and human proteins referenced in work from this lab (Melis et al. 2024) would not significantly accumulate in heterologous systems, e.g., cyanobacteria, when expressed by themselves, albeit under the control of a strong promoter (Bentley et al. 2013; Formighieri and Melis 2014; Chaves et al. 2017; Zhang et al. 2021; Majhi and Melis 2025a), evidently due to the cellular proteasome activity designed to degrade denatured and foreign proteins (Zhang et al. 2021). This recombinant protein stability problem is apparently alleviated by the “fusion constructs technology.”

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10811-026-03986-7>.

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Author contributions Bharat Kumar Majhi (BKM) and Anastasios Melis (AM) designed the project. BKM conducted the experimental work. AM quantified some of the results. BKM drafted the figures and wrote the first version of the article, which was edited by AM.

Data availability All data generated or analyzed in this work are included in this published version and the associated supplementary information files. Further data can be obtained from the corresponding author upon reasonable request.

Declarations

Competing interest Authors have no competing financial interests that could have influenced the work reported in this article.

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References

- Berla BM, Saha R, Immethun CM, Maranas CD, Moon TS, Pakrasi HB (2013) Synthetic biology of cyanobacteria: unique challenges and opportunities. *Front Microbiol* 4:246
- Bentley FK, García-Cerdán JG, Chen H-C, Melis A (2013) Paradigm of monoterpene (β -phellandrene) hydrocarbons production via photosynthesis in cyanobacteria. *Bioenergy Res* 6:917–929
- Betterle N, Hidalgo Martinez D, Melis A (2020) Cyanobacterial production of biopharmaceutical and biotherapeutic proteins. *Front Plant Sci* 11:237
- Chaves JE, Kirst H, Melis A (2015) Isoprene production in *Synechocystis* under alkaline and saline growth conditions. *J Appl Phycol* 27:1089–1097
- Chaves JE, Rueda-Romero P, Kirst PH, Melis A (2017) Engineering isoprene synthase expression and activity in cyanobacteria. *ACS Synth Biol* 6:2281–2292
- Domínguez-Martín MA, Sauer PV, Kirst H, Sutter M, Bina D, Greber BJ, Nogales E, Polívka T, Kerfeld CA (2022) Structures of a phycobilisome in light-harvesting and photoprotected states. *Nature* 609:835–845
- Formighieri C, Melis A (2014) Regulation of β -phellandrene synthase gene expression, recombinant protein accumulation, and monoterpene hydrocarbons production in *Synechocystis* transformants. *Planta* 240:309–324
- Formighieri C, Melis A (2015) A phycocyanin-phellandrene synthase fusion enhances recombinant protein expression and β -phellandrene (monoterpene) hydrocarbons production in *Synechocystis* (cyanobacteria). *Metab Eng* 32:116–124
- Formighieri C, Melis A (2016) Sustainable heterologous production of terpene hydrocarbons in cyanobacteria. *Photosynth Res* 130:123–135
- Hidalgo Martinez D, Betterle N, Melis A (2022) Phycocyanin fusion constructs for heterologous protein expression accumulate as functional heterohexameric complexes in cyanobacteria. *ACS Synth Biol* 11:1152–1166
- Hidalgo Martinez D, Melis A (2023) Cyanobacterial phycobilisomes as a platform for the stable production of heterologous enzymes and other proteins. *Metab Eng* 77:174–187
- Kefala A, Amprazi M, Mylonas E, Kotsifaki D, Providaki M, Pozidis C, Fotiadou M, Kokkinidis M (2021) Probing protein folding with sequence-reversed α -helical bundles. *Int J Mol Sci* 22:1955
- Khan MZ, Zugaza JL, Torres Aleman I (2025) The signaling landscape of insulin-like growth factor 1. *J Biol Chem* 301:108047
- Kirst H, Formighieri C, Melis A (2014) Maximizing photosynthetic efficiency and culture productivity in cyanobacteria upon minimizing the phycobilisome light-harvesting antenna size. *Biochim Biophys Acta Bioenerg* 1837:1653–1664
- Kodru S, ur Rehman A, Vass I (2020) Chloramphenicol enhances photosystem II photodamage in intact cells of the cyanobacterium *Synechocystis* PCC 6803. *Photosynth Res* 145:227–235
- Lacroix E, Viguera A, Serrano L (1998) Reading protein sequences backwards. *Folding Design* 3:79–85
- LeRoith D, Holly JMP, Forbes BE (2021) Insulin-like growth factors: ligands, binding proteins, and receptors. *Mol Metab* 52:101245
- Li J, Choi E, Yu H, Bai X-c (2019) Structural basis of the activation of type 1 insulin-like growth factor receptor. *Nat Commun* 10:4567

- Lindberg P, Park S, Melis A (2010) Engineering a platform for photosynthetic isoprene production in cyanobacteria, using *Synechocystis* as the model organism. *Metab Eng* 12:70–79
- Majhi BK, Melis A (2024) Recombinant protein synthesis and isolation of human interferon alpha-2 in cyanobacteria. *Bioresour Technol* 400:130664
- Majhi BK, Melis A (2025a) Engineering fibroblast growth factor-2 (FGF2) production in cyanobacteria. *ACS Synth Biol* 14:3978–3989
- Majhi BK, Melis A (2025b) Over-expression of a phycocyanin-interferon fusion and differential cleaving efficiency in cyanobacteria (*Synechocystis* sp. PCC 6803). *Biotechnol Bioeng* 123:752–762
- Maki RG (2010) Small is beautiful: insulin-like growth factors and their role in growth, development, and cancer. *J Clin Oncol* 28:4985–4995
- Melis A, Hidalgo Martinez DA, Betterle N (2024) Perspectives of cyanobacterial cell factories. *Photosynth Res* 162:459–471
- Nickle A, Ko S, Merrill AE (2024) Fibroblast growth factor 2. *Differentiation* 139:100733
- Ornitz DM, Itoh N (2001) Fibroblast growth factors. *Genome Biol* 2:reviews3005.1
- Panahi M, Cheng X, Alli Z, Sardana R, Callaghan M, Phipps J, Altosaar I (2003) Plant-derived recombinant human insulin-like growth factor precursor prohormone IGF-1B caused differentiation of human neuroblastoma cell lines SH-SY5Y. *Mol Breed* 12:21–31
- Park J, Yan G, Kwon KC, Liu M, Gonnella PA, Yang S, Daniell H (2020) Oral delivery of novel human IGF-1 bioencapsulated in lettuce cells promotes musculoskeletal cell proliferation, differentiation and diabetic fracture healing. *Biomaterials* 233:119591
- Poreba E, Durzynska J (2020) Nuclear localization and actions of the insulin-like growth factor 1 (IGF-1) system components: transcriptional regulation and DNA damage response. *Mutat Res Rev Mutat Res* 784:108307
- Poudel SB, Bhattarai G, Kook S-H, Shin Y-J, Kwon T-H, Lee S-Y, Lee J-C (2017) Recombinant human IGF-1 produced by transgenic plant cell suspension culture enhances new bone formation in calvarial defects. *Growth Horm IGF Res* 36:1–10
- Price S, Kuzhiumparambil U, Pernice M, Ralph PJ (2020) Cyanobacterial polyhydroxybutyrate for sustainable bioplastic production: critical review and perspectives. *J Environ Chem Eng* 8:104007
- Romand S, Jostock T, Fornaro M, Schmidt J, Ritter A, Wilms B, Laux H (2016) IGF-1 production in IGF-1R knockout CHO cells. *Biotechnol Bioeng* 113:1094–1101
- Rotwein P (2017) Diversification of the insulin-like growth factor 1 gene in mammals. *PLoS ONE* 12:e0189642
- Santos-Merino M, Yun L, Ducat DC (2023) Cyanobacteria as cell factories for the photosynthetic production of sucrose. *Front Microbiol* 14:1126032
- Singh NK, Sonani RR, Rastogi RP, Madamwar D (2015) The phycobilisomes: an early requisite for efficient photosynthesis in cyanobacteria. *EXCLI J* 14:268–289
- Singh RK, Tiwari SP, Rai AK, Mohapatra TM (2011) Cyanobacteria: an emerging source for drug discovery. *J Antibiot* 64:401–412
- Singh Y, Kaushal S, Sodhi RS (2020) Biogenic synthesis of silver nanoparticles using cyanobacterium *Leptolyngbya* sp. WUC 59 cell-free extract and their effects on bacterial growth and seed germination. *Nanoscale Adv* 2:3972–3982
- Tan S, Tan X, Chi Z, Dayin Zhang D, Li W (2018) In vitro assessment of the toxicity of lead (Pb²⁺) to phycocyanin. *Chemosphere* 192:171e177
- Vajdos FF, Ultsch M, Schaffer ML, Deshayes KD, Liu J, Skelton NJ, de Vos AM (2001) Crystal structure of human insulin-like growth factor-1: detergent binding inhibits binding protein interactions. *Biochemistry* 40:11022–11029
- Yao T, Asayama Y (2017) Animal-cell culture media: history, characteristics, and current issues. *Reprod Med Biol* 16:99–117
- Zhang X, Betterle N, Hidalgo Martinez D, Melis A (2021) Recombinant protein stability in cyanobacteria. *ACS Synth Biol* 10:810–825
- Zheng L, Zheng Z, Li X, Wang G, Zhang K, Wei P, Zhao J, Gao N (2021) Structural insight into the mechanism of energy transfer in cyanobacterial phycobilisomes. *Nat Commun* 12:5497

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